Random Variables:

1) Random variables are not Variables

2) mathematically, it is a function from the sample space to the real line

Toss a coin: $S = \{H, T\}$

$\downarrow 1/2 \quad \searrow 1/2$

R.V X $\quad X(H) = r \in \mathbb{R} \quad X(T) = t \in \mathbb{R}$

Throw a die: $S = \{1, 2, 3, 4, 5, 6\}$

$\downarrow 1/6 \quad \downarrow 1/6 \quad \downarrow 1/6 \quad \downarrow 1/6 \quad \downarrow 1/6 \quad \searrow 1/6$

$X(\omega)$ for all $\omega \in S$

ω	$X(\omega)$
1	r_1
2	r_2
3	r_3
4	r_4
5	r_5
6	r_6

$X(\omega) = 0 \rightarrow$ trivial random variable

$X(\omega) = c \rightarrow$ constant random variable

$X(\omega) = \omega \rightarrow$ identity random variable

ω	1	2	3	4	5	6
$X(\omega)$	0	1	2	0	1	2

[We can define events using random variables]

Is $X=0$ an event? yes: $\{\omega \in S : X(\omega)=0\}$
$= \{1,4\}$

$\Rightarrow P(X=0) = 2/6 = 1/3$

$$P(X=r) = \begin{cases} 1/3 & \text{if } r=0,1,2 \\ 0 & \text{else} \end{cases} \quad r \in \mathbb{R}$$

Discrete random variable: Takes a finite/ countable set of values

Probability mass function of a discrete random variable: for a random variable x, the PMF of x (denoted by $P_X(\cdot)$) is defined as $P_X(x) = P(X=x)$ $\quad P_X : \mathbb{R} \longrightarrow [0,1]$

Throw a die twice $\quad S = \{1,2,3,4,5,6\}^2$

x: number in 1st throw $\quad Z = 10x+y$

y: number in 2nd throw

$$P_X(x) = \begin{cases} 1/6 & \text{if } x=1,2,3,4,5,6 \\ 0 & \text{else} \end{cases}$$

$$P_Y(y) = \begin{cases} 1/6 & \text{if } x=1,2,3,4,5,6 \\ 0 & \text{else} \end{cases}$$

W = sum of two throws

$W \in \text{supp}(W) = \{2,3,4,5,6,7,8,9,10,11,12\}$

$P(W=2) = 1/36$

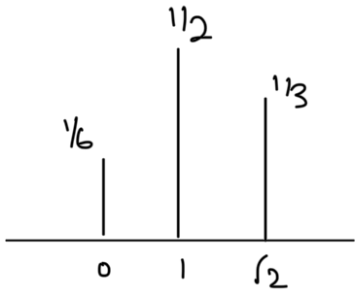
$$P(W=w) = \begin{cases} \frac{w-1}{36} & 2 \le w \le 7 \\ \frac{13-w}{36} & 8 \le w \le 12 \end{cases}$$
$w \in \text{supp}(w)$

PMF Properties:

1) $P_X(x) \ge 0$ $\quad P_X : \mathbb{R} \to [0,1]$

2) $\sum_{x \in \mathcal{X}} P_X(x) = 1$ $\quad \mathcal{X} = \text{supp}(P_X) = \{x : P_X(x) > 0\}$ ↓ (discrete set)

Valid PMF



1/6 1/2

S = {apple, orange, banana}

1/3

w	apple	orange	banana
$X(w)$	0	1	$\sqrt{2}$

Common discrete Random Variables:

1) Bernoulli Random Variable

any PMF $P_X(\cdot)$ corresponds to a distribution

$\mathcal{Q} \subseteq \mathbb{R} \Rightarrow \text{prob}(\mathcal{Q}) = \sum_{x \in \mathcal{Q} \cap \text{supp}(P_X)} P_X(x)$

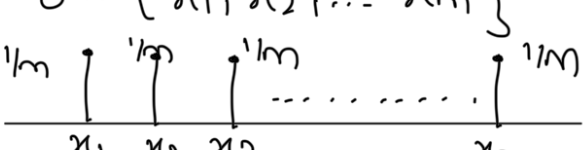
$X \sim \text{Bernoulli}(p)$ $\quad p \in [0,1]$ $\quad P_X(0) = 1-p$ $\quad P_X(1) = p$



2) Uniform on a discrete set

$X \sim \text{Uniform}(U)$ ← finite set

$U = \{x_1, x_2, \ldots x_m\}$



3) n independent Bernoulli (p)

$$X \sim \text{Binomial}(n,p)$$

X = number of successes (1)

$$\in \{0,1,2,\ldots,n\}$$

$$P_X(k) = {}^nC_1\, p^k (1-p)^{1-k}$$

$$\left(\frac{n}{k}\right)^k \le {}^nC_k \le \left(ne/k\right)^k$$

4) Geometric distribution

$$x \sim \text{Geometric}(p)$$

repeated independent bernoulli(p) trials

X = trials of 1st success $\in \{1,2,3,\ldots\}$

$P_X(k) = (1-p)^{k-1} p$ (peak near $k=1$)

5) Poisson's distribution:

$$x \sim \text{Poisson}(\lambda)$$

$x \in \{0,1,2,3,\ldots\}$ $\lambda \rightarrow$ rate of arrival

$$P_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

6) Continuous distribution:

→ assign probabilities to intervals

$$x \in [a,b]$$

Probability density function (PDF):

f_x: function from $\mathbb{R} \rightarrow \mathbb{R}$ such that

$$Pr(X \in [a,b]) = \int_a^b f_X(x)\,dx$$

$\rightarrow f_X(x) \geq 0$ in any interval

$\rightarrow \int_{supp(x)} f_X(x)\,dx = 1$ $\qquad supp(x) = \{x : f_X(x) > 0\}$

1) Uniform distribution:

$X \sim$ uniform $[a,b]$

$$f_X(x) = \begin{cases} \frac{1}{b-a}; & a \leq x \leq b \\ 0 & \text{else} \end{cases}$$

$$\text{probability of } [a,a] = \int_{a-}^{a+} f_X(x)\,dx = 0$$

$$Pr(X \in [a,b]) = Pr(x \in [a,b)) = Pr(x \in (a,b]) = Pr(x \in (a,b))$$

$f_X(x)$ of x in $[0,10]$

$\Rightarrow f_X(x) = cx \Rightarrow \int_0^{10} cx\,dx = 1 \Rightarrow c = 1/50$

Cumulative distributive function

R.V $X : S \rightarrow \mathbb{R}$

CDF of X, denoted by F_X, is a function from $\mathbb{R}$ to $[0,1]$, is defined as

$$F_X(x) = P(X \le x)$$

Event: $\{\omega \in S : X(\omega) < x\}$

Proper definition of PDF:

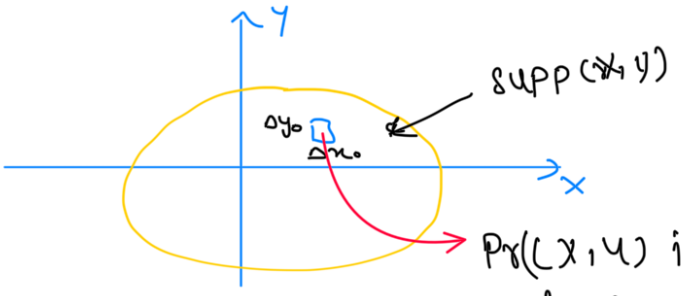
If there's an f_X such that the CDF $F_X(x) = \int_{-\infty}^{x} f_X(u)du$, then f_X is the PDF of X.

F_X is differential $\Rightarrow f_X = \frac{d}{dx} F_X$

Bivariate distributions:

$X, Y \sim$ PDF $f_{XY}(x,y)$ — $X, Y \to$ jointly continuous with PDF $f_{XY}(x,y)$

$\text{supp}(X,Y) = \{(x,y) : f_{XY}(x,y) > 0\}$



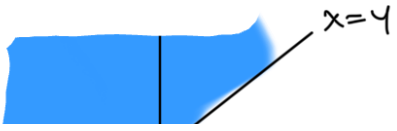
$\Pr((X,Y) \text{ in small region}) \approx f_{XY}(x_0, y_0) \cdot \Delta x_0 \cdot \Delta y_0$

$$f_{XY}(x,y) = \begin{cases} e^{-x-y} & ; x,y > 0 \\ 0 & \text{else} \end{cases}$$

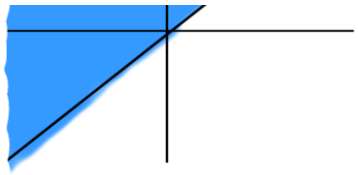
$\text{supp}(X,Y) =$ quadrant 1

Probability of a region: $X, Y \sim f_{XY}$

$\to P(X < Y)$



$$\int_{y=-\infty}^{\infty}\int_{x=-\infty}^{y} f_{XY}(x,y)\,dx\,dy$$



Marginal PDFs: $f_X(x) = \int_{y=-\infty}^{\infty} f_{XY}(x,y)\,dy$

$$f_Y(y) = \int_{x=-\infty}^{\infty} f_{XY}(x,y)\,dx$$

independance of random variables:

X&Y → Random variables jointly distributed

Events defined using X: $X \leq x \quad x \in \mathbb{R}$

$$\bigcup_{n=1}^{\infty} X \leq x - 1/n = X < x$$

Events defined using Y: $Y \leq y \quad y \in \mathbb{R}$

Independent RV's ⇒ X & Y are independent if

$P(X \leq x, Y \leq y) = P(X \leq x) \cdot P(Y \leq y) \quad \forall\, x, y$

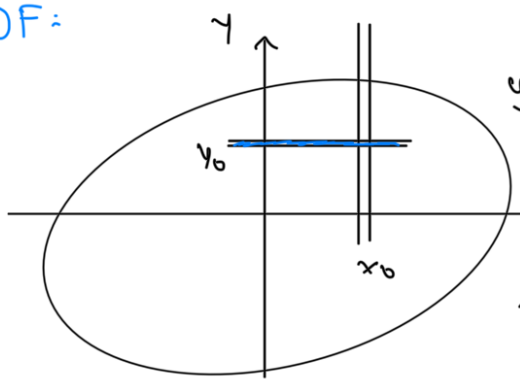
↓ joint CDF of X&Y ; ↓ CDF of X ; ↓ CDF of Y

$$f_{XY}(x,y) = \frac{d^2}{dx\,dy} F_{XY}(x,y)$$

↓ jointly continuous

discrete: $P_{XY}(X=x, Y=y) = P_X(X=x)\,P_Y(Y=y)$

Conditional PDF:



A) $y_0 \le y < y_0 + \Delta y_0$

B) $x_0 \le x < x_0 + \Delta x_0$

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{f_{XY}(x_0, y_0)\,\Delta x_0\,\Delta y_0}{f_X(x_0)\,\Delta x_0} = \boxed{\frac{f_{XY}(x_0, y_0)\,\Delta y_0}{f_X(x_0)}}$$

$P(Y \approx y_0 / X \approx x_0)$

conditional density

density of $Y/X = x_0$

$$\frac{f_{XY}(x_0, y_0)}{f_X(x_0)} = f_{Y/X=x_0}(y_0) \quad ; \quad f(x_0) > 0 \quad y : f_{XY}(x, y) > 0$$

(or)

$$f_{Y/X}(y_0 | x_0)$$

Discrete and continuous:

X: discrete PMF $P_X(x)$ $x \in X$

Y: continuous PDF $f_Y(y)$

$X \sim$ Uniform $\{-1, 1\}$

$Y/X = 1 \sim$ PDF $\frac{1}{\sqrt{2\pi}} e^{(y-1)^2/2}$

$Y/X = -1 \sim$ PDF $\frac{1}{\sqrt{2\pi}} e^{(y+1)^2/2}$

Bayes' rule and Total probability of RV's

discrete ≈ same as events

$X, Y \sim$ joint PMF p_{XY}

$$p_{XY}(x,y) = p_X(x)\, p_{Y/X=x}(y) = p_Y(y)\, p_{X/Y=y}(x$$

jointly continuous:

$X, Y \sim$ joint PMF f_{XY}

$$f_{XY}(x,y) = f_X(x)\, f_{Y/X=x}(y) = f_Y(y)\, f_{X/Y=y}(x$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\, dy = \int_{-\infty}^{\infty} f_{X/Y=y}(x)\, f_Y(y)\, dy$$